

Compelled Conclusions Over Model Classes

Beyond Knife-Edge Degeneracies

Micro Theory Lunch

May 5, 2026

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Problem

- ▶ Knife-edge counterexamples break universal claims.
- ▶ We need to distinguish fragile failure from major failure.

Main theorem

- ▶ Compulsion is equivalent to a measure-theoretic size notion.
- ▶ This allows use of geometric methods to prove compulsion.

Formalization

- ▶ We introduce **noise compulsion**
- ▶ It asks whether the conclusion survives small nondegenerate perturbations.

Consequences

- ▶ Planner's problem: Uniqueness of the solution is compelled.
- ▶ Potential games: equilibrium discreteness is compelled.

Motivation

Motivation

Knife-Edge Examples

Next part

Example 1: When Does Comparative Statics Behave Well?

Let $z(p, \theta)$ be the excess demand. For a fixed θ , define $z_\theta(p) := z(p, \theta)$. An equilibrium price p^* solves $z_\theta(p^*) = 0$.

- ▶ We are interested in **comparative statics** $\frac{\partial p^*}{\partial \theta}$.
- ▶ By **Implicit differentiation**:

$$0 = \frac{d}{d\theta} z(p^*(\theta), \theta) = \partial_p z(p^*(\theta), \theta) \frac{\partial p^*}{\partial \theta} + \partial_\theta z(p^*(\theta), \theta),$$

hence

$$\frac{\partial p^*}{\partial \theta} = -(\partial_p z)^{-1} \partial_\theta z.$$

- ▶ This only works if p^* is a **Regular root** of z_θ , i.e.

$$z_\theta(p^*) = 0 \quad \det(D_p z_\theta(p^*)) \neq 0$$

Bad cases are knife-edge

- ▶ **Heuristic:** almost all square matrices have non-zero determinant, so one expects irregular equilibria to be rare.
- ▶ **Caution:** this does *not* automatically transfer to a structured model class $\{z(\cdot, \theta) : \theta \in \Theta\}$.

We believe certain regularities of z would let us apply the heuristic and conclude irregular equilibria are knife-edge.

Next slide gives a concrete instance. We believe the same logic applies to a broad class of models.

A concrete instance

Let $P = (0, 2)$ and $\theta \in (-1, 1)$.

► **Demand:** $d(p) := 1 - (p - 1)^3$.

► **Supply:** $s(\theta) := 1 - \theta$.

► **Excess demand:**

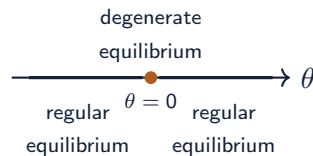
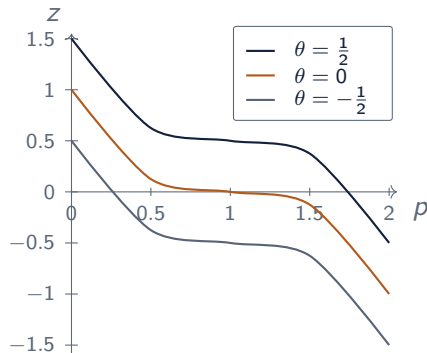
$$z(p, \theta) := d(p) - s(\theta) = \theta - (p - 1)^3.$$

Unique equilibrium:

$$z(p^*(\theta), \theta) = 0 \iff p^*(\theta) = 1 + \sqrt[3]{\theta} \in (0, 2).$$

$$D_p z(p, \theta) = -3(p-1)^2 \Rightarrow D_p z(p^*(\theta)) = -3\theta^{2/3}.$$

So equilibria are regular for every $\theta \neq 0$, but irregular at $\theta = 0$.



Example 2: A Knife-Edge Multiplicity Claim

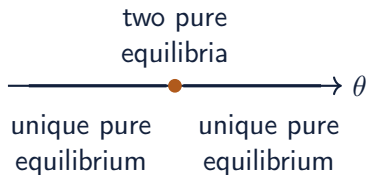
Consider the following entry game:

$$\theta \in (-1, 1)$$

$G(\theta) =$		E	O
	E	$(-\theta^2, -\theta^2)$	$(-1, 0)$
	O	$(0, -1)$	$(0, 0)$

- ▶ If $\theta \neq 0$, the unique pure equilibrium is (O, O) .
- ▶ If $\theta = 0$, both (O, O) and (E, E) are equilibria.

Any epistemic doubt about the exact equality $\theta = 0$ means allowing nearby values of θ ; then uniqueness prevails almost surely.



Motivation

What's Lacking

Next part

What We Want the Framework to Explain

Observation

Certain *bad behaviors* seem knife-edge.

- ▶ Degenerate/irregular equilibria (e.g. singular Jacobians).
- ▶ Non-uniqueness in optimization (multiple maximizers / flat objectives).
- ▶ Continuum of equilibria.

Plan. (1) Formally define the vague “knife-edge” intuition.

(2) Build a machinery that lets us prove and interpret such knife-edge claims.

Related Literature

Parallel Ideas Outside Economics

A theory restricted to universal statements (results holding everywhere) can be inexpressive: it cannot talk about typical behavior.

Probability

Many foundational results are a.s./a.e., not pointwise (e.g. SLLN, martingale convergence, ergodic theorems).

Dynamical systems

Many useful qualitative conclusions are generic (e.g. hyperbolic equilibria; Kupka–Smale-type genericity).

The broad idea is not new. We need our own notion of ‘typical’ motivated by economic applications.

Relevant Literature in Economics

- ▶ **Size based approach:** Exceptional/Bad set is “small”.
 - ▶ **Lebesgue a.e. (finite-dim):** Debreu [5], Dierker [6], Smale [21], Polemarchakis and Siconolfi [17].
 - ▶ **Prevalence/shyness (infinite-dim analogue):** Christensen [4], Hunt et al. [11], Anderson and Zame [1], Shannon and Zame [20], Heifetz and Neeman [10].
 - ▶ **Topological genericity (residual/meager):** Haller and Lagunoff [8], Chen and Xiong [3].
 - ▶ **Econometrics “almost every parameter” / generic identification:** Rothenberg [18], Sargan [19], McManus [14], Lewbel [13].
- ▶ **Perturbation based approach:** Speak to a model’s properties by looking at its perturbed variants. Small payoff or information perturbations can bring extra regularity. (Harsanyi [9], Kohlberg and Mertens [12], Carlsson and Van Damme [2], Morris and Shin [15], Frankel et al. [7]).

Knife-Edge: Size vs. Perturbations

- ▶ **Perturbation view:** Perturbation-based approaches are well motivated, but often tailored to a specific setting, so they do not yield a reusable criterion for arbitrary implications.
- ▶ **Size view:** Size-based approaches provide reusable machinery, but it is not obvious why “smallness” of the bad set is the right criterion, and different notions of smallness (topological vs measure theoretic) can disagree (Oxtoby [16], Chen and Xiong [3]).

To solve these issues, we connect the two views: we show that our perturbation-based notion of compulsion is equivalent to an appropriate measure-theoretic size statement.

How Size-Based Tools Fall Short

- ▶ **Model classes are proper subsets of ambient C^k :** genericity/prevalence tools often live on large function spaces (e.g. C^k), but in applications we care about a finite-dimensional model class; “small” in C^k need not translate to “small” within the parameterized family.
- ▶ **Model classes may not be linear subspaces of C^k :** many economically relevant families are constrained (e.g. convexity defines a cone), so we need tools that work beyond linear subspaces.
- ▶ **Domains can be non-open:** many workhorse arguments assume open domains, while economic models often live on compact or otherwise constrained sets where boundary effects matter.

We augment the machinery to handle these issues, making it more widely applicable.

Compulsion

Compulsion

Informal Idea

Next part

An Informal Definition

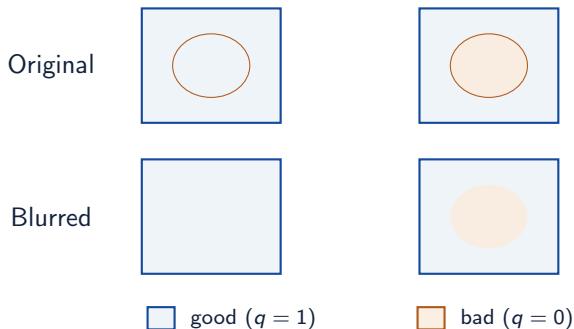
Call a counterexample *knife-edge* if the proposition is restored once we allow arbitrarily small local noise around the model.

A proposition $q : \Theta \rightarrow \{0, 1\}$ is **noise compelled** at θ if, after adding an arbitrarily small local blur η to θ , $\mathbb{E}[q(\theta + \eta)] = 1$.

This goes beyond pointwise failure by recording how substantial the failure set is, not just whether it is nonempty.

Graphical intuition

- ▶ Each noise η defines a blur operator $\theta \mapsto \mathbb{E}[q(\theta + \eta)]$
- ▶ q is compelled at θ if the blur operator maps θ to 1 for every small blur (made precise next)
- ▶ Blurring makes thin sets disappear but keeps thick ones.
- ▶ Compelled truths are those that survive squinting your eyes.



Why a Noise-Based Notion of Knife-Edge?

Principle

We use parameter perturbations to capture a variety of relevant economic drivers.

- ▶ **Epistemic doubt:** what we know usually identifies a neighborhood of models, not an exact point in parameter space.
- ▶ **Operational relevance:** theory is often used downstream with estimated or calibrated objects, so the exact parameter value is not what later users actually work with.
- ▶ **Numerical inaccuracies:** approximation, discretization, and computation replace the exact object by a nearby one even before any estimation enters.

Compulsion

Formal Definition

Next part

Ambient Space and Perturbations

Let $\mathcal{V}$ be a potentially infinite dimensional Banach space and let $\Theta \subset \mathcal{V}$ be open. We think of Θ as the parameter space.

nondegenerate V^ϵ -perturbation

Let $V \subset \mathcal{V}$ be a finite-dimensional subspace and $\epsilon > 0$. A **nondegenerate V^ϵ -perturbation** is a V -valued random variable η supported in $\overline{B_V(\epsilon)}$ whose law has density strictly positive a.e. on $B_V(\epsilon)$.

- ▶ We restrict to finite-dimensional V so “positive density on $B_V(\epsilon)$ ” is well-defined (via Lebesgue measure on V).
- ▶ Fix V, ϵ . If $\mathbb{E}[\tilde{q}(\theta + \eta)] = 1$ for one nondegenerate V^ϵ -noise η , then it holds for all such η (so checking one noise is enough).

Compulsion($\square$)

Let $q : \Theta \rightarrow \{0, 1\}$ be a proposition. Throughout, propositions are assumed Borel measurable.

Define the extension $\tilde{q} : \mathcal{V} \rightarrow \{0, 1\}$ by $\tilde{q}(\theta) := \begin{cases} q(\theta) & \text{if } \theta \in \Theta \\ 0 & \text{if } \theta \notin \Theta \end{cases}$.

Noise compulsion operator

We say q is V^ϵ noise compelled (or compelled when obvious from context) at θ , denoted by $\square_{V^\epsilon} q(\theta)$, if for every nondegenerate V^ϵ -perturbation η , $\mathbb{E}[\tilde{q}(\theta + \eta)] = 1$, equivalently, $\mathbb{P}(\tilde{q}(\theta + \eta) = 1) = 1$. When ϵ is dropped, $\square_V$ is taken to mean $\square_{V^\epsilon}$ for some $\epsilon > 0$.

- ▶ V identifies directions the noise acts in while ϵ identifies its size
- ▶ We abbreviate $\forall \theta \in \Theta : \square_{V^\epsilon} q(\theta)$ to $\square_{V^\epsilon} q$.

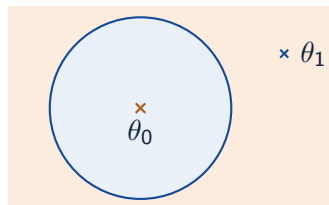
Pointwise Truth vs. Compulsion

Independence

$\Box_V q(\theta)$ and $q(\theta)$ are independent.

$$\Box_{\mathbb{R}^2} q(\theta_0) = 1, \quad q(\theta_0) = 0$$

$$\Box_{\mathbb{R}^2} q(\theta_1) = 0, \quad q(\theta_1) = 1$$



$\square q = 1$

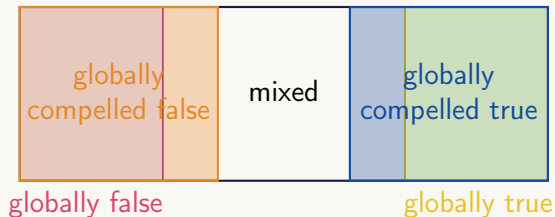
$\square q = 0$

Even though locally there is no relationship between the two, globally there is:

$$q \Rightarrow \Box_V q$$

How Compulsion Says More

Global



Pointwise truth gives three global cases. Global compulsion overlays a finer classification on top of them.

Local

At a point, pointwise truth still only says whether $q(\theta)$ is true or false.

Compulsion augments it with a notion of **stochastic truth near θ** .

Duality of Compulsion and Size

Duality of Compulsion and Size

Local Duality

Next part

Local Duality on a Fixed Slice

Define the true and false sets $T(q) := \{\theta \in \Theta : q(\theta) = 1\}$ and $F(q) := \{\theta \in \Theta : q(\theta) = 0\}$. Fix $\theta \in \Theta$ and $\epsilon > 0$ with $\theta + \overline{B_V(\epsilon)} \subseteq \Theta$. For measurable $S \subseteq \Theta$, define the directional density at scale ϵ by

$$D_{V^\epsilon}(S, \theta) := \frac{\lambda_V(\{v \in B_V(\epsilon) : \theta + v \in S\})}{\lambda_V(B_V(\epsilon))}.$$

Proposition: Compulsion as Full Directional Density

$$\square_{V^\epsilon} q(\theta) \iff D_{V^\epsilon}(T(q), \theta) = 1 \iff D_{V^\epsilon}(F(q), \theta) = 0$$

Duality of Compulsion and Size

Global Duality

Next part

From Local Duality to a Global Notion of Size

The local proposition tells us what “large” and “small” mean on a fixed slice. Can this be made global?

$$\forall \theta \in \Theta : \square_V q(\theta) \iff T(q) \text{ is large} \iff F(q) \text{ is small}$$

- ▶ In many economic applications, the parameter space is infinite-dimensional
- ▶ Lebesgue measure is trivial in infinite-dimensional spaces
- ▶ The correct extension of full/zero Lebesgue measure to infinite dimensions is **prevalence**/shyness

Finite Shyness / Prevalence Under a Probe

Fix a finite-dimensional probe $P \subseteq \mathcal{V}$ and write λ_P for Lebesgue measure on P .

Finite shyness (ambient)

A **Borel** set $S \subseteq \mathcal{V}$ is **finitely shy with probe** P if, for every $v \in \mathcal{V}$, the translate $S + v$ meets P in a λ_P -null set: $\lambda_P(P \cap S + v) = 0$. An arbitrary set is finitely shy with probe P if it is contained in a finitely shy Borel set.

Finite prevalence (ambient)

A set $A \subseteq \mathcal{V}$ is **finitely prevalent with probe** P if $\mathcal{V} \setminus A$ is finitely shy with probe P .

Relative version on $\Theta \subseteq \mathcal{V}$

A set $A \subseteq \Theta$ is **finitely prevalent in Θ with probe** P if $\Theta \setminus A$ is finitely shy with probe P .

Prevalence Is Not the Same as Genericity

Genericity

Topological: open dense / residual.

Prevalence

Measure-like: “almost every” along finite-dimensional probe slices.

- ▶ Genericity can behave badly from an “almost sure” perspective.
- ▶ A comeager set can still have Lebesgue measure 0.
- ▶ For this talk, prevalence is the right bridge from perturbations to negligible exceptions.

Proposition: Finite Prevalence with Probe V = Global V -Compulsion

$$\begin{aligned} & T(q) \text{ is finitely prevalent in } \Theta \text{ with probe } V \\ \iff & F(q) \text{ is finitely shy in } \Theta \text{ with probe } V \\ \iff & \forall \theta \in \Theta : \square_V q(\theta) \end{aligned}$$

- ▶ Prevalence gives the global size statement; compulsion gives the perturbational interpretation.
- ▶ Global probe enlargement *does* work (via duality + Fubini/Tonelli).

Appendix

Global Duality: Prevalence and Compulsion

Proof Sketch

For $\theta + \overline{B_V(\epsilon)} \subseteq \Theta$, local duality says:

$$\square_{V^\epsilon} q(\theta) \iff D_{V^\epsilon}(F(q), \theta) = 0.$$

- ▶ If $T(q)$ is finitely prevalent with probe V , then on every affine slice parallel to V the bad set is λ_V -null; in particular, it is null on a small V -ball around each $\theta \in \Theta$.
- ▶ By local duality, this implies $\square_V q(\theta)$ for every $\theta \in \Theta$.
- ▶ Conversely, if $\square_V q(\theta)$ holds for every θ , then each point on each affine V -slice has a neighborhood on which the bad set is λ_V -null.
- ▶ A countable subcover upgrades this local nullity to λ_V -nullity on the whole slice, hence finite prevalence.

Proof (Appendix)

Global Duality for Compatibility ($\Diamond$)

Define $\Diamond_V q(\theta) := \neg \Box_V (\neg q)(\theta)$.

Proposition: Compatibility and Failure of Shyness

$$\begin{aligned} & T(q) \text{ is *not* finitely shy in } \Theta \text{ with probe } V \\ \iff & F(q) \text{ is *not* finitely prevalent in } \Theta \text{ with probe } V \\ \iff & \exists \theta \in \Theta : \Diamond_V q(\theta) \end{aligned}$$

- ▶ Roughly: the true set fails to be null on *some* affine V -slice.
- ▶ This is a strict weakening of prevalence in general infinite dimensions.

Proof Sketch of the Global Duality for $\Diamond$

Sketch

Since $F(q) = T(\neg q)$, apply the global duality proposition to $\neg q$:

$$F(q) \text{ is finitely prevalent in } \Theta \text{ with probe } V \iff \forall \theta \in \Theta : \Box_V(\neg q)(\theta).$$

Negating yields

$$F(q) \text{ is *not* finitely prevalent} \iff \exists \theta \in \Theta : \neg \Box_V(\neg q)(\theta),$$

and the right-hand side is $\exists \theta : \Diamond_V q(\theta)$ by definition. The equivalence between $F(q)$ not prevalent and $T(q)$ not shy is by the definition of finite prevalence.

Proof (Appendix)

Proof Machinery

Proof Machinery

Transversality

Next part

Why Differential Geometry Helps

Full-measure and prevalence statements respond well to differential-geometric tools.

A basic example is Sard's theorem: if $f : M \rightarrow N$ is smooth, with M, N smooth finite-dimensional manifolds, then the set of critical values of f has measure 0 in N .

- ▶ Sard-type arguments control the size of the behavior of interest.
- ▶ Duality then turns this size statement into compulsion.

From Models to Objects We Can Study

Models are indexed by $\theta \in \Theta$.

We study them through associated objects encoding the relevant information about the model where M, N are finite-dimensional manifolds:

$$\mathcal{F}_\Theta = (f_\theta)_{\theta \in \Theta}, \quad f_\theta : M \rightarrow N.$$

- ▶ Equilibrium Analysis: payoffs, best responses, or first-order conditions.
- ▶ Mechanism design / constrained optimization: objectives, constraints, or KKT conditions.

Example: Unique Choice via Two-Point Inspection

Let X be open and $f_\theta : X \rightarrow \mathbb{R}$ a differentiable utility. A maximizer, if it exists, is unique if no distinct critical points have the same value.

For each $x \in X$, the jet map records the local data

$$j^1 f_\theta(x) = (x, f_\theta(x), Df_\theta(x)) \in J^1(X, \mathbb{R}) \quad j_2^1 f_\theta(x_1, x_2) = (j^1 f_\theta(x_1), j^1 f_\theta(x_2)) \in J_2^1(X, \mathbb{R})$$

Define distinct pairs as $\text{Conf}_2(X) := \{(x_1, x_2) \in X \times X : x_1 \neq x_2\}$.

$$\tau : \text{Conf}_2 \times \Theta \rightarrow J_2^1(X, \mathbb{R}) \quad \tau(x_1, x_2, \theta) := j_2^1 f_\theta(x_1, x_2)$$

We want to know how image of τ and $\tau_\theta := \tau(\cdot, \theta)$ crosses the following bad set:

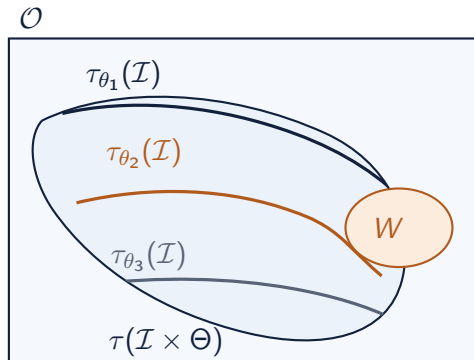
$$W = \{((x_1, y_1, p_1), (x_2, y_2, p_2)) : p_1 = 0, p_2 = 0, y_1 = y_2\}.$$

Inspection Spaces, Observations, and Bad Events

More generally, we study f_θ through finite-dimensional inspections.

- ▶ $\mathcal{I}$: an **inspection manifold** of finite-dimensional inspections, e.g. $\text{Conf}_2(X)$
- ▶ $\mathcal{O}$: an **observation manifold** of possible outcomes, e.g. $J_2^1(X, \mathbb{R})$
- ▶ $\tau : \mathcal{I} \times \Theta \rightarrow \mathcal{O}$: the **inspector map**, e.g. $\tau(x_1, x_2, \theta) = j_2^1 f_\theta(x_1, x_2)$
- ▶ $W \subseteq \mathcal{O}$: the observations we want to rule out.

For each θ , write $\tau_\theta := \tau(\cdot, \theta) : \mathcal{I} \rightarrow \mathcal{O}$ and study how $\tau_\theta(\mathcal{I}) \cap W$ (equivalently $\tau_\theta^{-1}(W) \subseteq \mathcal{I}$) behaves.



The plan for $\tau_\theta^{-1}(W)$ and $\tau_\theta(\mathcal{I}) \cap W$

- ▶ **Richness:** Submersion along inspection and finite-dimensional probe directions implies probe transversality.
- ▶ **Parametric Transversality:** Under the appendix's smoothness threshold, probe transversality gives $\tau_\theta \pitchfork W$ for finitely prevalent $\theta \in \Theta$.
- ▶ **Codimension** Transversality makes finitely prevalent $\tau_\theta^{-1}(W)$ a manifold of dimension $\dim(\mathcal{I}) - \text{codim}_\mathcal{O} W$.
- ▶ **Duality:** By global duality, this is equivalent to a compulsion statement about size of the bad set $\tau_\theta^{-1}(W)$.

Since we need the parametric transversality step to give finite prevalence, we need a finite-dimensional version of submersion and transversality.

Machinery defs (Appendix)

Definition

We call a family $\mathcal{F}_\Theta$ (τ, Q) -rich if τ is C^1 , $Q \subseteq \mathcal{V}$ is finite-dimensional, and at each (m, θ) ,

$$D\tau_{(m,\theta)}(T_m M \times Q) = T_{\tau(m,\theta)} N.$$

- ▶ In presence of richness, $\tau \pitchfork W$ for every submanifold $W \subseteq N$.
- ▶ This is the first step of the four-step argument discussed in the previous slide.

Proof Machinery

Verifying Richness

Next part

Splitting Parameters

Decompose $\mathcal{V}$ into vector spaces $\mathcal{V} = \mathcal{A} \oplus \mathcal{B}$ with $\mathcal{A}$ being finite dimensional.

Each $\theta \in \Theta$ can be written as $\theta = (\alpha, \beta)$ with $\alpha \in \mathcal{A}$ and $\beta \in \mathcal{B}$.

Take probe directions $Q := \mathcal{A} \times \{0\} \subseteq \mathcal{V}$ and define $\mathcal{P} \subset T(M \times \Theta)$ by

$$\mathcal{P}_{(m,\theta)} := T_m M \times Q.$$

Surjection From a subset of parameters is enough

If for every (m, α, β) the partial derivative

$$D_\alpha \tau_{(m,\alpha,\beta)} : \mathcal{A} \rightarrow T_{\tau(m,\alpha,\beta)} N$$

is surjective, then τ is a $\mathcal{P}$ -submersion, hence $\tau \pitchfork_{\mathcal{P}} W$ for every submanifold $W \subseteq N$.

Affine Families $\Rightarrow j^1$ -richness

For jet-based inspections, τ is typically a 1-jet inspector of the form

$$\tau(x, \theta) = j^1 f_\theta(x) \in J^1(X, Y).$$

Affine-in- $\mathcal{A}$ sufficient condition

Let $\mathcal{V} = \mathcal{A} \oplus \mathcal{B}$ with $\mathcal{A} = \mathbb{R} \times \mathbb{R}^d$ and $M = \mathbb{R}^d$. Suppose the jointly C^2 family $f_\theta : M \rightarrow \mathbb{R}$ has the form:

$$f_{((\alpha_0, \alpha_1), \beta)}(x) = g_\beta(x) + \alpha_0 + \alpha_1^\top x.$$

Parameter directions span the value and derivative coordinates; variations in x supply the base-point coordinates. Thus $D\tau(T_x M \times Q)$ spans the full jet tangent, giving (j^1, Q) -richness with $Q = \mathcal{A}$.

If instead the probe directions span cubics (Hermite interpolation), the same idea yields (j_2^1, Q) -richness.

j_2^1 (Appendix)

Applications

Applications

Regularities

Next part

Three consequences of richness

Setup

Let M be a d -dimensional manifold without boundary. Assume the stated smoothness and richness with probe Q .

- ▶ **Regular roots.** If $f : M \times \Theta \rightarrow \mathbb{R}^k$ is C^r with $r > \max(d - k, 0)$ and the family is (j^1, Q) -rich, then for finitely prevalent θ every root m with $f_\theta(m) = 0$ is regular (i.e. $Df_\theta(m)$ is surjective). Proof: regular roots
- ▶ **Morseness.** If $f : M \times \Theta \rightarrow \mathbb{R}$ is C^2 and the family is (j^1, Q) -rich, then for finitely prevalent θ the function f_θ is Morse; in particular $\text{Crit}(f_\theta)$ is discrete. Proof: Morseness
- ▶ **Distinct critical values.** If $\Phi : M \times \Theta \rightarrow \mathbb{R}$ is C^2 and the family is (j^1_2, Q) -rich, then for finitely prevalent θ distinct critical points have distinct values, hence a maximizer is unique (if it exists). Proof: unique maximizer

Applications

Economic Consequences

Next part

Applications

Assume open choice and strategy sets; equilibria are pure.

- ▶ **Regular equilibria / comparative statics.** If the square system $z(p, \theta)$ is jointly C^1 in p and probe parameters and (j^1, Q) -rich, then for finitely prevalent θ its equilibria are regular, with local differentiable comparative statics.
- ▶ **Planner's problem.** If the objective family Φ_θ is C^2 and (j_2^1, Q) -rich, then the maximizer is unique (if it exists) for finitely prevalent θ .
- ▶ **Potential games (potential-maximizing selection).** Pure equilibria of an exact potential game are critical points. If its C^2 potential family is (j^1, Q) -rich, equilibria are discrete for finitely prevalent θ . If moreover Φ_θ is (j_2^1, Q) -rich, distinct equilibria do not tie in Φ_θ , hence the potential-maximizing equilibrium is unique (if it exists).
- ▶ **Conjecture (general games and welfare; not proved here).** Once a welfare functional W_θ is specified from primitives, we suspect a two-point richness condition generically breaks welfare ties among equilibria, yielding a unique welfare-maximizing equilibrium (if it exists).

Conclusion

Conclusion

1. **Noise compulsion:** a claim holds almost surely under every sufficiently small nondegenerate perturbation along the chosen probe.
2. **Duality:** global compulsion $\iff$ finite prevalence (equivalently: the bad set is finitely shy), with the same probe.
3. **Tool:** by duality, we can prove compelled results by proving prevalence; differential geometry (transversality) plus richness delivers such prevalence statements.
4. **Workflow:** functional form $\Rightarrow$ richness $\Rightarrow$ finite prevalence $\Rightarrow$ compulsion (hence robust micro regularities).

Main message: a claim can fail at some models yet be compelled at every model; its failures are null on every affine slice along the chosen probe.

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Appendix

Appendix: Table of Contents

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Prevalence

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Appendix: Shyness and Prevalence

Let $\mathcal{V}$ be a complete metric vector space.

Transverse measure

For a Borel set $S \subseteq \mathcal{V}$, a Borel measure μ on $\mathcal{V}$ is *transverse* to S if:

1. there exists a compact set $K \subseteq \mathcal{V}$ such that $0 < \mu(K) < \infty$, and
2. for every $v \in \mathcal{V}$ one has $\mu(S + v) = 0$.

Shy / prevalent

A Borel set $S \subseteq \mathcal{V}$ is *shy* if there exists a measure transverse to S . An arbitrary set $A \subseteq \mathcal{V}$ is *shy* if it is contained in a shy Borel set. A set $T \subseteq \mathcal{V}$ is *prevalent* if its complement is shy.

Appendix: Prevalence/Shyness Under a Probe

Let $T \subseteq \mathcal{V}$.

Probe

A finite-dimensional subspace $P \subseteq \mathcal{V}$ is a *probe* for T if Lebesgue measure λ_P on P (viewed as a measure on $\mathcal{V}$ supported on P) is transverse to a Borel set containing T^c .

Slice viewpoint

If P is a probe for T , then for every $v \in \mathcal{V}$,

$$\lambda_P((T^c + v) \cap P) = 0.$$

Equivalently: for every $f \in \mathcal{V}$, for λ_P -a.e. $p \in P$ one has $f + p \in T$.

Appendix: Finite-dimensional Benchmark

Assume $\mathcal{V} = \mathbb{R}^d$.

Shyness $\leftrightarrow$ zero measure

For $S \subseteq \mathbb{R}^d$, S is shy with probe $\mathcal{V}$ if and only if S has Lebesgue measure 0.

Prevalence $\leftrightarrow$ full measure

For $T \subseteq \mathbb{R}^d$, T is prevalent in $\mathcal{V}$ (equivalently: finitely prevalent with probe $\mathcal{V}$) if and only if T has full Lebesgue measure.

Appendix

Local Calculus

Next part

Proposition: Compulsion as a Modal Operator

When $\Theta = \mathcal{V}$, for fixed V and ϵ , $\Box_{V^\epsilon}$ and $\Diamond_{V^\epsilon}$ are dual modal operators satisfying K -axioms.

For any open Θ , the radius-existential operator $\Box_V$ and its dual $\Diamond_V$ satisfy $K4$ -axioms.

Appendix **Logic of Compulsion** develops this by providing a local calculus for the operators.

Proposition: Local Calculus for $\Box_{V^\epsilon}(\cdot)(\theta)$ (and $\Diamond_{V^\epsilon}$)

Let $q_n : \Theta \rightarrow \{0, 1\}$, $n \in \mathbb{N}$, and assume $\theta + \overline{B_V(r)} \subseteq \Theta$ for every radius r below.

- ▶ **D (seriality):** $\Box_{V^\epsilon} q_1(\theta) \Rightarrow \Diamond_{V^\epsilon} q_1(\theta)$.
- ▶ **Countable conjunction:** if $\Box_{V^\epsilon} q_n(\theta)$ holds for all $n \in \mathbb{N}$, then $\Box_{V^\epsilon} (\bigwedge_{n \in \mathbb{N}} q_n)(\theta)$.
- ▶ **Countable disjunction:** if $\exists n \in \mathbb{N} : \Diamond_{V^\epsilon} q_n(\theta)$ then $\Diamond_{V^\epsilon} (\bigvee_{n \in \mathbb{N}} q_n)(\theta)$
- ▶ **Weakening:** $\Box_{V^\epsilon} q_1(\theta) \Rightarrow \Box_{V^\epsilon} (q_1 \vee q_2)(\theta)$.
- ▶ **Modus ponens:** $\Box_{V^\epsilon} (q_1 \Rightarrow q_2)(\theta) \wedge \Box_{V^\epsilon} q_1(\theta) \Rightarrow \Box_{V^\epsilon} q_2(\theta)$.
- ▶ **Mixed modus ponens:** $\Box_{V^\epsilon} (q_1 \Rightarrow q_2)(\theta) \wedge \Diamond_{V^\epsilon} q_1(\theta) \Rightarrow \Diamond_{V^\epsilon} q_2(\theta)$.
- ▶ **Chain implications:** $\Box_{V^\epsilon} (q_1 \Rightarrow q_2)(\theta) \wedge \Box_{V^\epsilon} (q_2 \Rightarrow q_3)(\theta) \Rightarrow \Box_{V^\epsilon} (q_1 \Rightarrow q_3)(\theta)$.
- ▶ **Monotonicity in ϵ :** if $0 < \epsilon' \leq \epsilon$ and $\Box_{V^\epsilon} q_1(\theta)$ holds, then $\Box_{V^{\epsilon'}} q_1(\theta)$.
- ▶ **Two-step (radius additivity):** $\Box_{V^{\epsilon_1}} (\Box_{V^{\epsilon_2}} q_1)(\theta) \iff \Box_{V^{\epsilon_1 + \epsilon_2}} q_1(\theta)$.

Corollary: Modal Logic Formulation

Modal axioms

- ▶ K consists of propositional logic plus:

$$(MP) \quad q_1, (q_1 \Rightarrow q_2) \Rightarrow q_2, \quad \Box(q_1 \Rightarrow q_2)(\theta) \Rightarrow (\Box q_1(\theta) \Rightarrow \Box q_2(\theta)),$$

$$(Nec) \quad (\forall \vartheta \in \Theta : q_1(\vartheta) = 1) \Rightarrow (\forall \vartheta \in \Theta : \Box q_1(\vartheta) = 1).$$

- ▶ $K4$ is K plus:

$$(4) \quad \Box q_1(\theta) \Rightarrow \Box(\Box q_1)(\theta).$$

Corollary

Fix a probe V and $\theta \in \Theta$.

- ▶ If $\Theta = \mathcal{V}$, then for each fixed $\epsilon > 0$, $\Box_{V^\epsilon}$ satisfies K .
- ▶ With the convention $\Box_V := \exists \epsilon > 0 \Box_{V^\epsilon}$, the operator $\Box_V(\cdot)(\theta)$ satisfies $K4$.

Local Probe Enlargement Can Fail

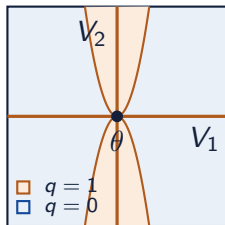
Even if a statement is compelled along one direction, enlarging the probe can introduce new failure directions. To enlarge locally, one needs $\Box_V$ to hold *uniformly* around θ , as formalized by the uniform tube lemma below.

Warning (local)

Local compulsion at a point is not stable under probe enlargement:

$$\Box_{V_1} q(\theta) \wedge \Box_{V_2} q(\theta) \not\Rightarrow \Box_{V_1+V_2} q(\theta).$$

$$\forall v_2 \in B_{V_2}(\epsilon) : \Box_{V_1} q(\theta + v_2) \not\Rightarrow \Box_{V_1+V_2} q(\theta).$$



Uniform Tube $\Rightarrow$ Larger-Probe Compulsion (Local)

Local probe enlargement is *not* automatic, but it becomes valid under a uniform lower bound in transverse directions.

Proposition: Uniform Tube Lemma

Let $V_1, V_2 \subset \mathcal{V}$ be finite-dimensional subspaces with $V_1 \cap V_2 = \{0\}$ and let $q : \Theta \rightarrow \{0, 1\}$. Fix $\theta \in \Theta$ and $\epsilon > 0$ with $\theta + B_{V_2}(\epsilon) \subseteq \Theta$. If

$$\forall v_2 \in B_{V_2}(\epsilon) : \quad \square_{V_1^\epsilon} q(\theta + v_2),$$

then $\square_{V_1 + V_2} q(\theta)$.

Proof (Appendix)

Appendix

Global Calculus

Next part

Global $\Box$ -Calculus

For each proposition $q_n : \Theta \rightarrow \{0, 1\}$, we write $\Box_V q_n$ as shorthand for $\forall \theta \in \Theta : \Box_V q_n(\theta)$. If Θ is finite-dimensional, canonically we take $\mathcal{V}$ to be the ambient direction space; probe enlargement is then an operation on finite-dimensional subspaces of $\mathcal{V}$.

Proposition: Global Probe Calculus (Not Pointwise)

Let $(q_n)_{n \in \mathbb{N}} : \Theta \rightarrow \{0, 1\}$, and let $V, V_1, V_2 \subset \mathcal{V}$ be finite-dimensional probes.

- ▶ **Necessitation:** if $q_1(\theta) = 1$ for all $\theta \in \Theta$, then $\Box_V q_1$.
- ▶ **Probe enlargement:** if $\Box_V q_1$ and $V' \supseteq V$ is finite-dimensional, then $\Box_{V'} q_1$.
- ▶ **Unifying probes:** if $\Box_{V_1} q_1$ and $\Box_{V_2} q_2$, then $\Box_{V_1 + V_2} (q_1 \wedge q_2)$.
- ▶ **Cross-probe modus ponens:** if $\Box_{V_1} (q_1 \Rightarrow q_2)$ and $\Box_{V_2} q_1$, then $\Box_{V_1 + V_2} q_2$.

Proof (Appendix)

Proposition: Global Probe Enlargement

Let $V_1 \subseteq V_2$ be finite dimensional subspace of $\mathcal{V}$. If $\square_{V_1} q$ holds, then $\square_{V_2} q$ holds.

- ▶ By global duality, $\square_{V_1} q$ means $(F(q) + \nu) \cap V_1$ has λ_{V_1} -measure 0 for every $\nu \in \mathcal{V}$.
- ▶ By Fubini/Tonelli (on $V_2 \cong V_1 \oplus (V_2/V_1)$), this implies $(F(q) + \nu) \cap V_2$ has λ_{V_2} -measure 0 for every ν , hence $\square_{V_2} q$.

Appendix

Machinery

Next part

Appendix: What Kind of Intersection Is Helpful?

Definition (Transversality; and the stronger P -version)

We say f is **transverse** to W , written $f \pitchfork W$, if for every $x \in M$ with $f(x) \in W$,

$$Df_x(T_x M) + T_{f(x)} W = T_{f(x)} N.$$

More strongly, if $P \subset TM$ is a finite-rank smooth subbundle, we say f is **P -transverse** to W , written $f \pitchfork_P W$, if for every $x \in M$ with $f(x) \in W$,

$$Df_x(P_x) + T_{f(x)} W = T_{f(x)} N.$$

- ▶ This has two key consequences:
 - ▶ **Preimage theorem:** if $f \pitchfork W$, then $f^{-1}(W)$ is a submanifold of M of codimension $\text{codim}_N W$.
 - ▶ **Parametric transversality:** for finite-dimensional U and $F \in C^r$ with $r > \max(\dim M - \text{codim } W, 0)$, $F \pitchfork W$ implies $F(\cdot, u) \pitchfork W$ for almost every $u \in U$.

Appendix: Sufficient Conditions for Transversality

Let $f : M \rightarrow N$ be C^1 and let $W \subseteq N$ be a smooth submanifold.

Definition (Submersion; and the stronger P -version)

We say f is a **submersion** if for every $x \in M$,

$$Df_x(T_x M) = T_{f(x)} N.$$

If $P \subset TM$ is a finite-rank smooth subbundle, we say f is a **P -submersion** if for every $x \in M$,

$$Df_x(P_x) = T_{f(x)} N.$$

Sufficiency for P -Transversality

If f is a submersion, then $f \pitchfork W$ for every submanifold $W \subseteq N$. If f is a P -submersion, then $f \pitchfork_P W$ for every submanifold $W \subseteq N$.

Appendix: Prevalence Parametric Transversality

Let Θ be open in a vector space $\mathcal{V}$. Since $\Theta \subseteq \mathcal{V}$, we identify $T_\theta\Theta \cong \mathcal{V}$ and fix a finite-dimensional linear subspace $Q \leq \mathcal{V}$ (independent of θ).

Proposition (Prevalence parametric transversality)

Let M, N be smooth finite-dimensional manifolds, let $W \subset N$ be a smooth submanifold, and let $F : M \times \Theta \rightarrow N$ be C^r with

$$r > \max(\dim M - \operatorname{codim} W, 0).$$

Let $\mathcal{P} \subset T(M \times \Theta)$ be the subbundle with fiber

$$\mathcal{P}_{(m,\theta)} := T_m M \times Q.$$

If $F \pitchfork_{\mathcal{P}} W$, then $F_\theta \pitchfork W$ for *finitely prevalent* $\theta \in \Theta$ with probe Q .

Appendix: Richness: Finite Parameter Directions That Suffice

Let Θ be open in a vector space $\mathcal{V}$, let $\tau : M \times \Theta \rightarrow N$ be a map, and let $W \subseteq N$ be a smooth submanifold.

Definition $((\tau, Q)$ -richness)

Fix a finite-dimensional subspace $Q \subseteq \mathcal{V}$ and define the domain-direction subbundle $\mathcal{P} \subset T(M \times \Theta)$ by

$$\mathcal{P}_{(m,\theta)} := T_m M \times Q.$$

We say τ is (τ, Q) -rich if $\tau \in C^1(M \times \Theta, N)$ and τ is a $\mathcal{P}$ -submersion, i.e. for every (m, θ) ,

$$D\tau_{(m,\theta)}(\mathcal{P}_{(m,\theta)}) = T_{\tau(m,\theta)} N.$$

Remark

If τ is (τ, Q) -rich, then $\tau \pitchfork_{\mathcal{P}} W$ (indeed $\tau \pitchfork_{\mathcal{P}} W'$ for every submanifold $W' \subseteq N$).

Appendix: Hermite interpolation $\Rightarrow j_2^1$ -richness

For two-point inspections, the inspector map is typically

$$\tau(x_1, x_2, \theta) = j_2^1 f_\theta(x_1, x_2) \in J_2^1(X, \mathbb{R}), \quad (x_1, x_2) \in \text{Conf}_2(X).$$

Cubic-in- $\mathcal{A}$ sufficient condition

Let $X \subset \mathbb{R}^d$ be open and let $\mathcal{V} = \mathcal{A} \oplus \mathcal{B}$ where $\mathcal{A}$ is the coefficient space of polynomials of total degree ≤ 3 . Suppose the jointly C^2 family is

$$f_{(\alpha, \beta)}(x) = h_\beta(x) + g_\alpha(x)$$

with g_α ranging over those cubics. Then for every $(x_1, x_2) \in \text{Conf}_2(X)$ the map

$$\alpha \longmapsto (g_\alpha(x_1), Dg_\alpha(x_1), g_\alpha(x_2), Dg_\alpha(x_2))$$

is a submersion onto the jet fibers. Variations in (x_1, x_2) supply the base-point coordinates, giving (j_2^1, Q) -richness with $Q = \mathcal{A}$.

Appendix

Proofs

Next part

Appendix: Proof of Local Duality for $\square$

Fix $\theta \in \Theta$ and $\epsilon > 0$ with $\theta + \overline{B_V(\epsilon)} \subseteq \Theta$. Let

$$A := \{v \in B_V(\epsilon) : \theta + v \in F(q)\}.$$

By definition, $\square_{V^\epsilon} q(\theta)$ means that for every nondegenerate V^ϵ -perturbation η ,

$$\mathbb{P}(\eta \in A) = \mathbb{P}(q(\theta + \eta) = 0) = 0.$$

($\Rightarrow$) If $\square_{V^\epsilon} q(\theta)$ holds, apply it to η uniform on $B_V(\epsilon)$. Then $\mathbb{P}(\eta \in A) = \lambda_V(A)/\lambda_V(B_V(\epsilon)) = 0$, so $D_{V^\epsilon}(F(q), \theta) = 0$.

($\Leftarrow$) If $D_{V^\epsilon}(F(q), \theta) = 0$, then $\lambda_V(A) = 0$. For any nondegenerate η with density f strictly positive a.e. on $B_V(\epsilon)$,

$$\mathbb{P}(\eta \in A) = \int_A f(v) d\lambda_V(v) = 0,$$

hence $\square_{V^\epsilon} q(\theta)$.

Since $T(q)$ and $F(q)$ partition the slice, $D_{V^\epsilon}(T(q), \theta) = 1 - D_{V^\epsilon}(F(q), \theta)$.

Appendix: Proof of Local Duality for $\Diamond$

Fix $\theta \in \Theta$ and $\epsilon > 0$ with $\theta + \overline{B_V(\epsilon)} \subseteq \Theta$. Recall $\Diamond_{V^\epsilon} q(\theta) := \neg \Box_{V^\epsilon}(\neg q)(\theta)$. By Local Duality for $\Box$ applied to $\neg q$,

$$\Box_{V^\epsilon}(\neg q)(\theta) \iff D_{V^\epsilon}(T(\neg q), \theta) = 1 \iff D_{V^\epsilon}(F(q), \theta) = 1,$$

hence

$$\Diamond_{V^\epsilon} q(\theta) \iff D_{V^\epsilon}(F(q), \theta) < 1.$$

Since $T(q)$ and $F(q)$ partition the slice, $D_{V^\epsilon}(T(q), \theta) = 1 - D_{V^\epsilon}(F(q), \theta)$, so

$$\Diamond_{V^\epsilon} q(\theta) \iff D_{V^\epsilon}(T(q), \theta) > 0.$$

Finally,

$$\begin{aligned} & \Diamond_{V^\epsilon} q(\theta) \wedge \Diamond_{V^\epsilon}(\neg q)(\theta) \\ \iff & D_{V^\epsilon}(T(q), \theta) > 0 \wedge D_{V^\epsilon}(F(q), \theta) > 0 \\ \iff & D_{V^\epsilon}(T(q), \theta) \in (0, 1). \end{aligned}$$

Appendix: Proof of Global Duality I

Recall: $\square_V q(\theta)$ means $\square_{V^\epsilon} q(\theta)$ for some $\epsilon > 0$.

($\Rightarrow$) Assume $T(q)$ is finitely prevalent in Θ with probe V , i.e. $F(q)$ is finitely shy in Θ with probe V . Then for every $\theta \in \Theta$,

$$\lambda_V((F(q) - \theta) \cap V) = 0.$$

Since Θ is open, choose $\epsilon > 0$ with $\theta + \overline{B_V(\epsilon)} \subseteq \Theta$; then also

$$\lambda_V((F(q) - \theta) \cap B_V(\epsilon)) = 0,$$

and by Local Duality $\square_{V^\epsilon} q(\theta)$ holds; thus $\square_V q(\theta)$ holds for every θ .

Appendix: Proof of Global Duality II

($\Leftarrow$) Assume $\forall \theta \in \Theta : \square_V q(\theta)$. Fix $\theta_0 \in \Theta$ and consider the V -slice $\Theta \cap (\theta_0 + V)$. At each ϑ in this slice, choose a witnessing radius $\epsilon_\vartheta > 0$ whose closed V -ball lies in Θ . By Local Duality,

$$\lambda_V((F(q) - \vartheta) \cap B_V(\epsilon_\vartheta)) = 0.$$

The sets $\vartheta + B_V(\epsilon_\vartheta)$ form an open cover of the slice, so by second countability of V there is a countable subcover $\{\vartheta_n + B_V(\epsilon_{\vartheta_n})\}_{n \in \mathbb{N}}$. For each n , set $t_n := \vartheta_n - \theta_0 \in V$. For any $u \in (F(q) - \theta_0) \cap V$, we have $\theta_0 + u \in F(q) \cap (\theta_0 + V)$. Pick n such that $\theta_0 + u \in \vartheta_n + B_V(\epsilon_{\vartheta_n})$, i.e. $\theta_0 + u = \vartheta_n + w$ for some $w \in B_V(\epsilon_{\vartheta_n})$. Then $u = t_n + w$ and also $w = (\theta_0 + u) - \vartheta_n \in (F(q) - \vartheta_n) \cap B_V(\epsilon_{\vartheta_n})$. Hence

$$(F(q) - \theta_0) \cap V \subseteq \bigcup_{n \in \mathbb{N}} \left(t_n + ((F(q) - \vartheta_n) \cap B_V(\epsilon_{\vartheta_n})) \right).$$

Appendix: Proof of Global Duality III

Each set inside the union is a translate of a λ_V -null set, hence λ_V -null; therefore the union is λ_V -null, and $\lambda_V((F(q) - \theta_0) \cap V) = 0$. Since θ_0 was arbitrary, $F(q)$ is finitely shy in Θ with probe V , i.e. $T(q)$ is finitely prevalent in Θ with probe V .

Appendix: Proof of Global Duality for $\Diamond$

Define $\Diamond_V q(\theta) := \neg \Box_V(\neg q)(\theta)$. Since $F(q) = T(\neg q)$, applying Global Duality for $\Box$ to $\neg q$ gives

$$F(q) \text{ is finitely prevalent in } \Theta \text{ with probe } V \iff \forall \theta \in \Theta : \Box_V(\neg q)(\theta).$$

Negating both sides yields

$$\begin{aligned} & F(q) \text{ is *not* finitely prevalent in } \Theta \text{ with probe } V \\ \iff & \exists \theta \in \Theta : \neg \Box_V(\neg q)(\theta) \\ \iff & \exists \theta \in \Theta : \Diamond_V q(\theta). \end{aligned}$$

Finally, $F(q)$ is finitely prevalent if and only if $T(q) = \Theta \setminus F(q)$ is finitely shy, so

$$F(q) \text{ not finitely prevalent} \iff T(q) \text{ not finitely shy.}$$

Appendix: Proof of Prevalence Parametric Transversality

Fix $\theta_0 \in \Theta$ and define the open set $U_{\theta_0} := \{\alpha \in Q : \theta_0 + \alpha \in \Theta\} \subseteq Q$. Define $\tilde{F} : M \times U_{\theta_0} \rightarrow N$ by $\tilde{F}(m, \alpha) = F(m, \theta_0 + \alpha)$. Since $\mathcal{P}_{(m, \theta)} = T_m M \times Q$, the hypothesis $F \pitchfork_{\mathcal{P}} W$ implies $\tilde{F} \pitchfork W$.

By the classical parametric transversality theorem (finite-dimensional parameter manifold Q ; $r > \max(\dim M - \operatorname{codim} W, 0)$), the set

$$E_{\theta_0} := \{\alpha \in U_{\theta_0} : \tilde{F}(\cdot, \alpha) \not\pitchfork W\}$$

is λ_Q -null. Equivalently, on the affine slice $\theta_0 + Q$ the exceptional set inside $\Theta \cap (\theta_0 + Q)$ is λ_Q -null, which is exactly finite prevalence with probe Q .

Appendix: Proof of j^1 -richness $\Rightarrow$ regular roots

Let $\tau(m, \theta) = j^1 f_\theta(m)$ and let $\pi : J^1(M, \mathbb{R}^k) \rightarrow \mathbb{R}^k$ be projection to the value coordinate. Then $f = \pi \circ \tau$ as a map $M \times \Theta \rightarrow \mathbb{R}^k$.

If the family is (j^1, Q) -rich, then τ is a $\mathcal{P}$ -submersion for $\mathcal{P} = TM \times Q$, hence f is also a $\mathcal{P}$ -submersion (since π is a surjective submersion). In particular $f \pitchfork_{\mathcal{P}} \{0\}$.

Since f is C^r with $r > \max(d - k, 0)$, prevalence parametric transversality (with probe Q) applies to f and $W = \{0\}$, giving: for finitely prevalent θ , $f_\theta \pitchfork \{0\}$. But $f_\theta \pitchfork \{0\}$ is exactly that 0 is a regular value of f_θ .

Appendix: Proof of j^1 -richness $\Rightarrow$ prevalent Morseness

Let $\tau(m, \theta) = j^1 f_\theta(m)$ and let $W_{\text{crit}} = \{(m, y, p) \in J^1(M, \mathbb{R}) : p = 0\}$. If the family is (j^1, Q) -rich, then τ is a $\mathcal{P}$ -submersion for $\mathcal{P} = TM \times Q$, hence $\tau \pitchfork_{\mathcal{P}} W_{\text{crit}}$. Apply prevalence parametric transversality (with probe Q) to (τ, W_{crit}) : for finitely prevalent θ , $\tau_\theta \pitchfork W_{\text{crit}}$. Unwinding the definition, this is exactly the Morse nondegeneracy condition at all critical points, hence f_θ is Morse.

Appendix: Proof of j_2^1 -richness $\Rightarrow$ unique potential maximizer (if exists)

Let M be a d -dimensional manifold and let $\Phi : M \times \Theta \rightarrow \mathbb{R}$ be C^2 . Define the two-point inspector

$$\tau : \text{Conf}_2(M) \times \Theta \rightarrow J_2^1(M, \mathbb{R}), \quad \tau(m_1, m_2, \theta) := j_2^1 \Phi_\theta(m_1, m_2).$$

Consider the submanifold

$$W_{\text{tie}} := \{((m_1, y_1, p_1), (m_2, y_2, p_2)) \in J_2^1(M, \mathbb{R}) : p_1 = 0, p_2 = 0, y_1 = y_2\}.$$

It has codimension $2d + 1$ in $J_2^1(M, \mathbb{R})$ (two d -dimensional gradient constraints and one scalar equality).

If the family is (j_2^1, Q) -rich, then τ is a $\mathcal{P}$ -submersion for $\mathcal{P} = T \text{Conf}_2(M) \times Q$, hence $\tau \pitchfork_{\mathcal{P}} W_{\text{tie}}$. By prevalence parametric transversality, for finitely prevalent θ one has $\tau_\theta \pitchfork W_{\text{tie}}$. Therefore $\tau_\theta^{-1}(W_{\text{tie}}) \subseteq \text{Conf}_2(M)$ is a submanifold of codimension $2d + 1$, but $\dim \text{Conf}_2(M) = 2d$, so necessarily $\tau_\theta^{-1}(W_{\text{tie}}) = \emptyset$.

Thus for such θ there do not exist distinct $m_1 \neq m_2$ with $D\Phi_\theta(m_1) = D\Phi_\theta(m_2) = 0$ and $\Phi_\theta(m_1) = \Phi_\theta(m_2)$. In particular, if Φ_θ has a potential-maximizing equilibrium among its critical points, it is unique.

Appendix: Proof of Local $\square_{V^\epsilon}$ -Calculus I

Fix a finite-dimensional probe V , fix $\epsilon > 0$, and fix $\theta \in \Theta$ with $\theta + \overline{B_V(\epsilon)} \subseteq \Theta$. Define $\diamond_{V^\epsilon} q(\theta) := \neg \square_{V^\epsilon}(\neg q)(\theta)$. Recall: $\square_{V^\epsilon} q(\theta)$ means that for every nondegenerate V^ϵ -perturbation η ,

$$\mathbb{P}(q(\theta + \eta) = 1) = 1.$$

For the $\square_{V^\epsilon}(\cdot)(\theta)$ bullets below, fix an arbitrary nondegenerate V^ϵ -perturbation η .

- ▶ **D:** if $\square_{V^\epsilon} q_1(\theta)$ held but $\diamond_{V^\epsilon} q_1(\theta)$ failed, then $\square_{V^\epsilon}(\neg q_1)(\theta)$ would hold by definition of $\diamond$, hence $\mathbb{P}(q_1(\theta + \eta) = 1) = 1$ and $\mathbb{P}(q_1(\theta + \eta) = 0) = 1$, a contradiction.
- ▶ **Countable conjunction:** if $q_n(\theta + \eta) = 1$ a.s. for each $n \in \mathbb{N}$, then $\mathbb{P}(\forall n : q_n(\theta + \eta) = 1) = 1$ since a countable intersection of probability-1 events has probability 1.

Appendix: Proof of Local $\Box_{V^\epsilon}$ -Calculus II

- **Countable disjunction:** for events $A_n := \{q_n(\theta + \eta) = 1\}$,

$$\mathbb{P}\left(\bigcup_{n \in \mathbb{N}} A_n\right) > 0 \iff \exists n \in \mathbb{N} : \mathbb{P}(A_n) > 0,$$

since if $\mathbb{P}(A_n) = 0$ for all n then $\mathbb{P}(\bigcup_n A_n) \leq \sum_n \mathbb{P}(A_n) = 0$. Therefore

$$\Diamond_{V^\epsilon}\left(\bigvee_{n \in \mathbb{N}} q_n\right)(\theta) \iff \exists n \in \mathbb{N} : \Diamond_{V^\epsilon} q_n(\theta),$$

because $\Diamond_{V^\epsilon} r(\theta)$ means that there exists some nondegenerate V^ϵ -perturbation η with $\mathbb{P}(r(\theta + \eta) = 1) > 0$.

- **Weakening:** if $q_1(\theta + \eta) = 1$ a.s., then $(q_1 \vee q_2)(\theta + \eta) = 1$ a.s.

Appendix: Proof of Local $\Box_{V^\epsilon}$ -Calculus III

- ▶ **Modus ponens:** if $(q_1 \Rightarrow q_2)(\theta + \eta) = 1$ a.s. and $q_1(\theta + \eta) = 1$ a.s., then $q_2(\theta + \eta) = 1$ a.s.

Pointwise justification: for each outcome where $(q_1 \Rightarrow q_2)$ and q_1 both evaluate to 1, classical propositional logic yields $q_2 = 1$, so the a.s. implication follows.

- ▶ **Mixed modus ponens:** if $(q_1 \Rightarrow q_2)(\theta + \eta) = 1$ a.s., then $\{q_1(\theta + \eta) = 1\} \subseteq \{q_2(\theta + \eta) = 1\}$ modulo a null set. Thus if $\mathbb{P}(q_1(\theta + \eta) = 1) > 0$, then $\mathbb{P}(q_2(\theta + \eta) = 1) > 0$, so $\Diamond_{V^\epsilon} q_2(\theta)$ holds.

- ▶ **Chain:** if $(q_1 \Rightarrow q_2)(\theta + \eta) = 1$ a.s. and $(q_2 \Rightarrow q_3)(\theta + \eta) = 1$ a.s., then $(q_1 \Rightarrow q_3)(\theta + \eta) = 1$ a.s.

Pointwise justification: for each outcome, $(q_1 \Rightarrow q_2)$ and $(q_2 \Rightarrow q_3)$ together imply $(q_1 \Rightarrow q_3)$ by propositional logic, giving the a.s. statement.

Appendix: Proof of Local $\square_{V^\epsilon}$ -Calculus IV

- **Monotonicity in ϵ :** if $0 < \epsilon' \leq \epsilon$ and $\square_{V^\epsilon} q_1(\theta)$ holds, let

$$A := \{v \in B_V(\epsilon) : q_1(\theta + v) = 0\}.$$

Taking η to be uniform on $B_V(\epsilon)$ gives $\lambda_V(A) = 0$. Hence $A \cap B_V(\epsilon')$ is λ_V -null, so for any nondegenerate $V^{\epsilon'}$ -perturbation η' we have $\mathbb{P}(q_1(\theta + \eta') = 0) = 0$, i.e. $\square_{V^{\epsilon'}} q_1(\theta)$.

- **Two-step (radius additivity):** fix $\epsilon_1, \epsilon_2 > 0$ with $\theta + \overline{B_V(\epsilon_1 + \epsilon_2)} \subseteq \Theta$, and write $r := \epsilon_1 + \epsilon_2$.

Appendix: Proof of Local $\square_{V^\epsilon}$ -Calculus V

($\Rightarrow$) Assume $\square_{V^{\epsilon_1}}(\square_{V^{\epsilon_2}} q_1)(\theta)$. By Local Duality, it suffices to show that the bad set

$$A := \{u \in B_V(r) : q_1(\theta + u) = 0\}$$

is λ_V -null. For $v \in B_V(\epsilon_1)$ define $A_v := \{w \in B_V(\epsilon_2) : v + w \in A\}$. For almost every v , outer compulsion gives $\square_{V^{\epsilon_2}} q_1(\theta + v)$, hence $\lambda_V(A_v) = 0$ by Local Duality. Therefore, by Fubini/Tonelli,

$$\int_{B_V(\epsilon_1)} \lambda_V(A_v) dv = \int_{B_V(\epsilon_1)} \int_{B_V(\epsilon_2)} 1_A(v + w) dw dv = 0,$$

which implies $1_A(v + w) = 0$ for $\lambda_V \times \lambda_V$ -a.e. $(v, w) \in B_V(\epsilon_1) \times B_V(\epsilon_2)$. Now for $u \in B_V(r)$,

$$(1_{B_V(\epsilon_1)} * 1_{B_V(\epsilon_2)})(u) = \lambda_V(B_V(\epsilon_1) \cap (u - B_V(\epsilon_2))) > 0,$$

Appendix: Proof of Local $\square_{V^\epsilon}$ -Calculus VI

and thus

$$\begin{aligned} 0 &= \int_{B_V(\epsilon_1)} \int_{B_V(\epsilon_2)} 1_A(v+w) dw dv \\ &= \int_{B_V(r)} 1_A(u) (1_{B_V(\epsilon_1)} * 1_{B_V(\epsilon_2)})(u) du \end{aligned}$$

forces $\lambda_V(A) = 0$. Hence $\square_{V^r} q_1(\theta)$.

($\Leftarrow$) Assume $\square_{V^r} q_1(\theta)$. By Local Duality, the bad set

$$A := \{u \in B_V(r) : q_1(\theta + u) = 0\}$$

is λ_V -null. For $v \in B_V(\epsilon_1)$ define $A_v := \{w \in B_V(\epsilon_2) : v + w \in A\}$. By Fubini/Tonelli,

$$\begin{aligned} \int_{B_V(\epsilon_1)} \lambda_V(A_v) dv &= \int_{B_V(\epsilon_1)} \int_{B_V(\epsilon_2)} 1_A(v+w) dw dv \\ &= \int_{B_V(r)} 1_A(u) (1_{B_V(\epsilon_1)} * 1_{B_V(\epsilon_2)})(u) du = 0, \end{aligned}$$

Appendix: Proof of Local $\square_{V^\epsilon}$ -Calculus VII

so $\lambda_V(A_v) = 0$ for λ_V -a.e. v . Therefore $\square_{V^{\epsilon_2}} q_1(\theta + v)$ holds for λ_V -a.e. $v \in B_V(\epsilon_1)$. Since any nondegenerate V^{ϵ_1} -perturbation has density > 0 a.e. on $B_V(\epsilon_1)$, it follows that $\square_{V^{\epsilon_1}} (\square_{V^{\epsilon_2}} q_1)(\theta)$ holds.

Since η was arbitrary, the $\square_{V^\epsilon}(\cdot)(\theta)$ bullets follow. The $\diamond_{V^\epsilon}$ bullets follow by the definition $\diamond_{V^\epsilon} r(\theta) := \neg \square_{V^\epsilon}(\neg r)(\theta)$ together with the two $\diamond$ bullets above. The last two bullets prove monotonicity and two-step additivity.

Appendix: Proof of Global Probe Calculus

For each $q_n : \Theta \rightarrow \{0, 1\}$, recall $\Box_V q_n$ abbreviates $\forall \theta : \Box_V q_n(\theta)$.

- ▶ **Necessitation:** if $q_1(\theta) = 1$ for all θ , then $\Box_V q_1(\theta)$ holds for all θ by definition, hence $\Box_V q_1$.
- ▶ **Probe enlargement:** by Global Duality, $\Box_V q_1$ implies $T(q_1)$ is finitely prevalent with probe V , i.e. for all $f \in \mathcal{V}$,

$$\lambda_V((F(q_1) + f) \cap V) = 0.$$

If $V' \supseteq V$, write $V' = V \oplus W$. For fixed f , define $E := (F(q_1) + f) \cap V'$, where $F(q_1) = \Theta \setminus T(q_1)$. For each $w \in W$, the V -section satisfies

$$E_w := \{v \in V : (v, w) \in E\} = ((F(q_1) + f - w) \cap V),$$

hence $\lambda_V(E_w) = 0$ for every w . By Fubini/Tonelli on $V \times W \cong V'$, $\lambda_{V'}(E) = 0$, so $T(q_1)$ is finitely prevalent with probe V' , hence $\Box_{V'} q_1$.

- ▶ **Unifying probes:** if $\Box_{V_1} q_1$ and $\Box_{V_2} q_2$, probe enlargement gives $\Box_{V_1+V_2} q_1$ and $\Box_{V_1+V_2} q_2$. Apply local conjunction pointwise to get $\Box_{V_1+V_2}(q_1 \wedge q_2)$.
- ▶ **Cross-probe modus ponens:** if $\Box_{V_1}(q_1 \Rightarrow q_2)$ and $\Box_{V_2} q_1$, enlarge both probes to $V_1 + V_2$ and apply local modus ponens pointwise to get $\Box_{V_1+V_2} q_2$.

Appendix: Proof of Uniform Tube Lemma

Assume $\forall v_2 \in B_{V_2}(\epsilon) : \square_{V_1^\epsilon} q(\theta + v_2)$. Let η be any nondegenerate $(V_1 + V_2)^\rho$ -perturbation for some $\rho > 0$ to be chosen.

Since $V_1 \cap V_2 = \{0\}$, the linear map $\Phi : V_1 \times V_2 \rightarrow V_1 + V_2$ given by $\Phi(v_1, v_2) = v_1 + v_2$ is a linear isomorphism. Hence Φ^{-1} is continuous (all norms are equivalent in finite dimension). Choose $\rho > 0$ so that

$$\Phi^{-1}(B_{V_1+V_2}(\rho)) \subseteq B_{V_1}(\epsilon) \times B_{V_2}(\epsilon).$$

Define the bad set in product coordinates:

$$A := \{(v_1, v_2) \in B_{V_1}(\epsilon) \times B_{V_2}(\epsilon) : \tilde{q}(\theta + v_1 + v_2) = 0\}.$$

For each $v_2 \in B_{V_2}(\epsilon)$, the hypothesis $\square_{V_1^\epsilon} q(\theta + v_2)$ implies the V_1 -section A_{v_2} is λ_{V_1} -null. By Fubini/Tonelli, $\lambda_{V_1 \times V_2}(A) = 0$.

Now set $S := \Phi(A) \cap B_{V_1+V_2}(\rho)$. Since Φ is linear (hence Lipschitz), $\lambda_{V_1+V_2}(\Phi(A)) = 0$, hence $\lambda_{V_1+V_2}(S) = 0$. Because η has a density strictly positive a.e. on $B_{V_1+V_2}(\rho)$, it follows that $\mathbb{P}(\eta \in S) = 0$. Therefore $\mathbb{P}(\tilde{q}(\theta + \eta) = 1) = 1$. Since η was arbitrary, $\square_{(V_1+V_2)^\rho} q(\theta)$ holds, i.e. $\square_{V_1+V_2} q(\theta)$.